

EMPLOYEE REQUESTS INSTALLATION OF LICENSED SOFTWARE THROUGH THE SERVICE CATALOG

Complete Project Documentation

ServiceNow – Service Portal, Service Catalog, Flow Designer, Approvals, Business Rules, ACLs and Notifications

Project Information	Details
Project Name	Software Installation Request Automation in ServiceNow
Platform	ServiceNow
Primary Interface	Service Portal / Service Catalog
Workflow	Catalog → Request/RITM → Approval → Catalog Task → Fulfilment → Status → Notification
Project Type	Service Request Automation
Documentation	Functional and Technical Project Documentation

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Purpose: To provide a structured, automated and trackable process for requesting installation of licensed software.

1. Executive Summary

The Software Installation Request Automation project focuses on developing a structured and automated process for employees who require licensed software. The solution uses ServiceNow Service Catalog and Service Portal to provide a simple request interface, while workflow automation manages validation, approval, fulfilment and notifications.

The objective is to reduce manual coordination between employees, managers, software support teams and administrators. Instead of relying on emails or informal communication, each software request is recorded as a ServiceNow request and Requested Item (RITM), processed through the configured approval flow, and converted into a fulfilment task after approval.

The project also considers licensing, IT governance and security requirements. Software name, version and business justification are treated as important request information. Requested For can be automatically populated, while department and location can be obtained dynamically from the user's profile.

The implementation is organized into five phases: Requirement Analysis & Planning; Backend Development & Configurations; UI/UX Development & Customization; Data Migration, Testing & Security; and Deployment, Documentation & Final Presentation.

Project Workflow

Employee → Service Catalog Item → Request/RITM → Validation → Approval → Software Support Task → Installation/Fulfilment → Status Update → Notification

Key Outcomes

- Centralized software installation request submission.
- Standardized capture of software and business-justification information.
- Automated approval routing.
- Automated creation of fulfilment tasks for the Software Support Team.
- Improved visibility and tracking of request status.
- Reduced manual coordination and data-entry errors.
- Better support for licensing, IT governance and security controls.

2. Business Objective and Scope

2.1 Business Objective

The business objective is to streamline and automate the end-to-end lifecycle of licensed software installation requests using ServiceNow. The solution is intended to improve operational efficiency, reduce delays, standardize approval and fulfilment, and provide transparent tracking.

- Reduce manual effort and eliminate unnecessary coordination delays.
- Accelerate approval and fulfilment of software installation requests.
- Enforce standardized workflows and dynamic approval processes.
- Provide centralized visibility into request progress.
- Enhance end-user experience through faster and transparent service delivery.
- Support compliance with software licensing, IT governance and security policies.

2.2 Functional Scope

The functional scope covers the automation of software installation requests in ServiceNow, including the Service Catalog item, variables, approvals, fulfilment tasks, notifications, validation, security and testing.

- Create a Software Installation Request Service Catalog item.
- Configure software-related catalog variables.
- Make critical information mandatory.
- Automatically populate Requested For.
- Support dynamic Department and Location information.
- Create Request and Requested Item records.
- Route requests for manager/security approval.
- Create a Software Support fulfilment task after approval.
- Track request and task states.
- Generate notifications at appropriate lifecycle stages.
- Apply access controls and validate data integrity.

3. Stakeholder Analysis

The project involves multiple stakeholders. Each stakeholder has a defined responsibility and expectation from the automated software request process.

Stakeholder	Role	Needs / Expectations	Impact of Automation
End Users / Requesters	Employees requesting software	Easy request form, quick approval, installation visibility and status tracking	Faster software access and transparent tracking
IT Support / Fulfilment Team	Installs approved software on devices	Complete request data, automated task assignment and less manual coordination	Standardized intake, fewer errors and more time for critical tasks
Approvers	Managers / Security Officers validating requests	Policy enforcement and structured approval process	Automated routing and better audit trail
ServiceNow Administrators	Configure catalog, workflows, rules and notifications	Flexible, maintainable and scalable automation	Easier maintenance and configurable workflows
IT Governance / Licensing	Ensures software use follows policy	Business justification, license availability and controlled installation	Improved compliance visibility

3.1 Stakeholder Responsibilities

- Requester submits complete and accurate software information.
- Approver reviews business need and compliance considerations.
- Software Support Team performs the approved installation task.
- ServiceNow Administrator maintains the platform configuration.
- Governance or licensing stakeholders can validate licensing requirements where applicable.

4. Phase 1 – Requirement Analysis & Planning

Phase 1 establishes the problem statement, objectives, stakeholders, functional requirements and implementation roadmap before configuration begins.

4.1 Step 1: Understand the Project Requirement

- Identify the problem: licensed software requests require a structured and controlled process.
- Decide to automate the software request lifecycle using ServiceNow.
- Define the main objective of reducing manual effort and improving request visibility.
- Identify the information employees must provide for a software installation request.
- Define the overall lifecycle: Request → Validation → Approval → Software Task → Fulfilment → Status Update → Notification.

4.2 Required Request Information

Information	Purpose	Validation / Behaviour
Software Name	Identifies the requested application	Mandatory
Software Version	Identifies the required version	Mandatory
Business Justification	Explains why the software is required	Mandatory
Requested For	Identifies the employee receiving the software	Auto-populated where configured
Department	Provides organizational context	Pulled from user profile where configured
Location	Provides user/device context	Pulled from profile where configured
Device / Installation Details	Supports fulfilment	Captured through catalog variables
License Availability / User Indicator	Supports licensing awareness	Tooltip/help text or validation as configured

5. Phase 1 – ServiceNow Components and Roadmap

5.1 Define the ServiceNow Components

- Service Catalog.
- Software Installation Request Catalog Item.
- Catalog Variables and Variable Set.
- Request (sc_request).
- Requested Item (sc_req_item).
- Catalog Task (sc_task).
- Approval records (sysapproval_approver).
- Flow Designer / Workflow automation.
- Business Rules.
- Catalog UI Policies and Catalog Client Scripts.
- Notifications.

5.2 Project Roadmap

The project follows a structured five-phase roadmap:

1. Phase 1: Requirement Analysis & Planning.
2. Phase 2: Backend Development & Configurations.
3. Phase 3: UI/UX Development & Customization.
4. Phase 4: Data Migration, Testing & Security.
5. Phase 5: Deployment, Documentation & Final Presentation.

Roadmap: Requirement Analysis → Backend Configuration → UI/UX Customization → Automation & Security → Testing → Documentation & Deployment

5.3 Requirements Summary

Requirement Type	Requirement
Functional	Employee can submit a software installation request through Service Catalog.
Functional	System generates REQ/RITM records.
Functional	Request is routed for approval.
Functional	Approved request creates a fulfilment task.
Functional	Notifications communicate lifecycle changes.
Security	Only authorized users can access or update relevant records.
Data	Mandatory fields prevent incomplete requests.
Usability	Form provides clear fields, help text and simple navigation.

6. Phase 2 – Backend Development & Configurations

Phase 2 establishes the backend data structure and supporting records required to process licensed software installation requests.

6.1 Data Architecture

The implementation uses the standard ServiceNow request structure in which a catalog submission creates a Request and Requested Item. The Requested Item represents the catalog request and carries the variables entered by the requester. After approval, a Catalog Task is created for the Software Support Team.

Record / Component	Purpose	Relationship
Service Catalog Item	Entry point for the employee request	Creates REQ/RITM
sc_request (REQ)	Stores the overall request	Parent-level request record
sc_req_item (RITM)	Represents the requested software item	Linked to REQ
sysapproval_approver	Stores approval information	Linked to approval lifecycle
sc_task (SCTASK)	Represents fulfilment work	Created after required approval
Notification	Communicates lifecycle changes	Triggered by configured conditions

6.2 Backend Design Principle

The backend is designed around a controlled progression of records. The catalog captures user input, the request records preserve the transaction, approval controls authorization, the task assigns fulfilment work, and notifications communicate the result.

7. Phase 2 – Request, RITM and Task Handling

7.1 Request Record

When the employee submits the catalog item, ServiceNow generates a Request record. The request number provides a central reference for tracking.

7.2 Requested Item (RITM)

The Requested Item represents the specific software installation item selected by the requester. Catalog variables entered on the form are available to the RITM and can be used by the automation.

7.3 Catalog Task

After approval, the automation creates a Catalog Task for the Software Support Team. The task contains the fulfilment information needed to install the requested software and update the task state.

Stage	Expected Record / Output
Submission	REQ and RITM created
Approval	Approval record generated
Approval Granted	Catalog Task created
Fulfilment	Task moves through Open / Work in Progress / Completed as configured
Completion	Request status is updated and notification is generated

Request, RITM and Catalog Task records:

The screenshot displays the ServiceNow Request form for Request ID REQ0010010. The form is titled "Request - REQ0010010" and includes a navigation bar with "All", "Favorites", "History", and "Admin" tabs. The form fields are organized into two columns. The left column contains fields for "Number" (REQ0010010), "Requested for" (System Administrator), "Location", "Due date" (2026-08-31 23:19:51), "Price" (\$0.00), "Description", "Short description", "Special instructions", and "Comments (Customer visible)". The right column contains fields for "Opened" (2026-08-29 23:19:51), "Opened by" (System Administrator), "Approval" (Approved), and "Request state" (Approved). The "Request state" field is highlighted with a green border. At the bottom right, there is a "Work notes" checkbox and a "Post" button.

8. Phase 2 – Approval Configuration

8.1 Approval Objective

Approval ensures that a software installation request is reviewed before fulfilment. The configured approval process can involve a manager and, where required, a security or licensing approver.

8.2 Approval Lifecycle

1. Employee submits the software request.
2. ServiceNow creates the REQ/RITM.
3. Flow Designer evaluates the request and starts the approval step.
4. Configured approver reviews the software, version and business justification.
5. Approver selects Approve or Reject.
6. Approved requests continue to fulfilment.
7. Rejected requests stop or follow the configured rejection path and notify the requester.

8.3 Approval Data

Field / Information	Purpose
Approver	Identifies the person responsible for the decision
Approval State	Shows Requested, Approved or Rejected state as configured
Request Reference	Connects approval to the related request
Decision Time	Provides lifecycle traceability

Approval record:

The screenshot shows the ServiceNow interface for a 'Requested Item' (RITM0010010). The form includes fields for Number, Due date, Approval (set to 'Approved'), Quantity (1), and Stage (Assess or Scope Task). Below these fields is a 'Delivery Plan Status' section showing a table of tasks. The tasks are 'Assess or Scope Task' and 'Provide requested service', both with a duration of 1 day and expected start dates of 2026-08-30 and 2026-08-31 respectively. The 'Assess or Scope Task' is marked as 'Completed'. At the bottom, there is a 'Description' field with a rich text editor containing the text: 'Allows employees to request software installations. fulfills via software support team.'

Task	Duration	Expected Start	Started	Completed
Assess or Scope Task	1 Day	2026-08-30	2026-08-30	
Provide requested service	1 Day	2026-08-31		

9. Phase 2 – Business Rules

Business Rules are used to automate backend logic when records meet configured conditions. For the software installation project, backend logic can be used to support status handling, licence-shortage exception processing, task assignment and completion behaviour.

9.1 Example Business Rule – Licence Unavailable

A licence-shortage condition can be used to identify a request that cannot currently be fulfilled. The supplied project configuration uses an On Hold condition and creates an incident for follow-up by Software Support.

Example logic from the project configuration:

```
if (current.state == 3) { // On Hold
  var inc = new GlideRecord('incident');
  inc.initialize();
  inc.short_description = 'License unavailable for ' + current.request_item;
  inc.description = 'Software request could not be fulfilled due to license shortage.';
  inc.caller_id = current.request.requested_for;
  inc.assignment_group.setDisplayValue('Software Support');
  inc.priority = 3;
  inc.insert();
}
```

9.2 Purpose of the Rule

- Detect a configured On Hold condition.
- Create an exception record for license shortage.
- Provide a clear description for support follow-up.
- Route the issue to the Software Support group.
- Preserve traceability between the request and exception handling.

10. Phase 3 – UI/UX Development & Customization

Phase 3 focuses on creating a simple and user-friendly Service Portal experience for employees requesting licensed software.

10.1 Interface Design Objective

The objective is to design a clear Software Installation Request catalog item using ServiceNow catalog variables and variable sets. The form should guide the employee to provide the information needed for approval and fulfilment.

10.2 Creation of Service Catalog Item

Navigation: ServiceNow → Service Catalog → Catalog Definitions → Maintain Items

1. Open Maintain Items.
2. Click New.
3. Create the Software Installation Request catalog item.
4. Enter the required item details.
5. Configure the item availability and other required settings.
6. Save the catalog item.

Output: Software Installation Request catalog item created successfully.

Software Installation Request Catalog Item:

The screenshot displays the ServiceNow interface for creating a new catalog item. The left sidebar shows the navigation menu with 'Maintain Items' selected. The main form area is titled 'Catalog Item - Software Installation Request' and includes the following fields and sections:

- Name:** Software Installation Request
- Application:** Global
- Active:** ☒
- Category:** Software
- Fulfillment automation level:** Unspecified
- State:** -- None --
- Checked out:** -- None --
- Owner:** System Administrator

Below the form fields, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Item Details' tab is active, showing a 'Short description' field with the text 'Request to install company-approved software.' and a 'Description' field with the text 'Allows employees to request software installations. Fulfills via Software Support Team.'

11. Phase 3 – Variables Configuration

Catalog variables collect the information required to process the software installation request. Critical fields are made mandatory to protect data quality.

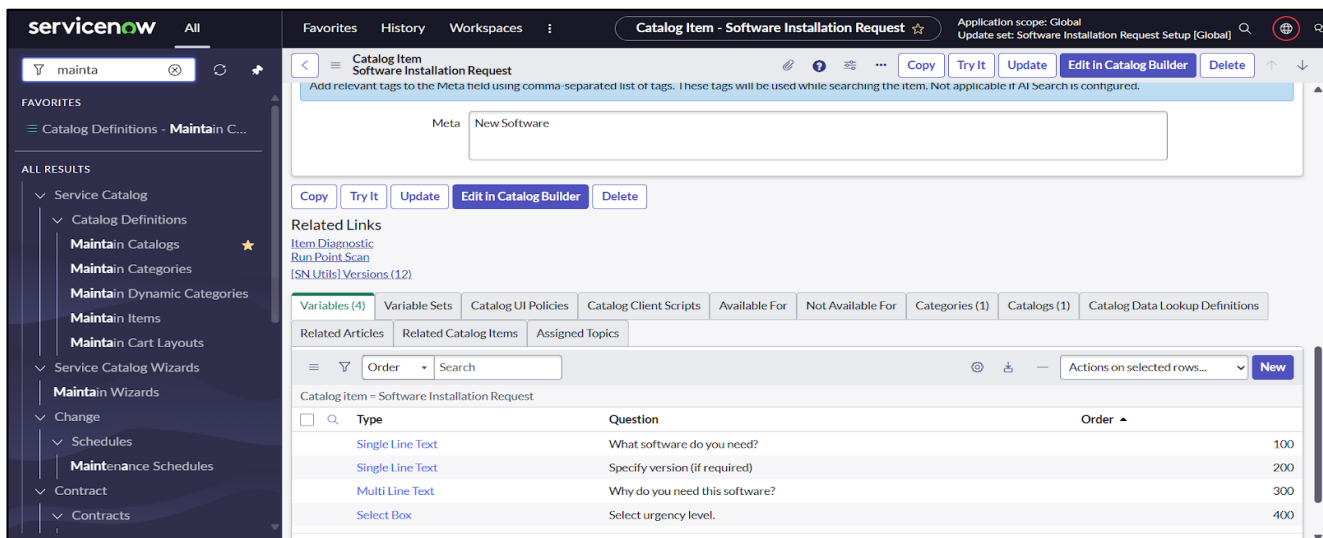
11.1 Variable Configuration Procedure

1. Open the Software Installation Request catalog item.
2. Navigate to Variables.
3. Create or edit the required variables.
4. Set mandatory fields where required.
5. Add choices for fields that require controlled values and Save it.

11.2 Recommended Variable Set

Variable	Example / Description	Required
Software Name	Name of the licensed software	Yes
Software Version	Required version	Yes
Business Justification	Reason the employee needs the software	Yes
Requested For	Employee receiving the software	Auto-populated / controlled
Department	Employee department	Auto-populated where configured
Location	Employee location	Auto-populated where configured
Device Details	Device information needed for installation	As configured
Additional Information	Supporting details	Optional where appropriate

Catalog variables:



The screenshot shows the ServiceNow Catalog Builder interface for a 'Software Installation Request' catalog item. The 'Variables' tab is selected, showing a table of variables to be configured. The table has columns for 'Type' and 'Question'. The variables listed are:

Type	Question	Order
Single Line Text	What software do you need?	100
Single Line Text	Specify version (if required)	200
Multi Line Text	Why do you need this software?	300
Select Box	Select urgency level.	400

12. Phase 3 – Variable Sets and Dynamic Behaviour

12.1 Variable Set Configuration

A Variable Set groups related variables so that the catalog item remains organized and easier to maintain.

1. Navigate to Service Catalog → Variable Sets.
2. Create the required Variable Set.
3. Add related software request variables.
4. Associate the Variable Set with the Software Installation Request catalog item.
5. Save the configuration.

Output: Software-related variables are organized using the configured Variable Set.

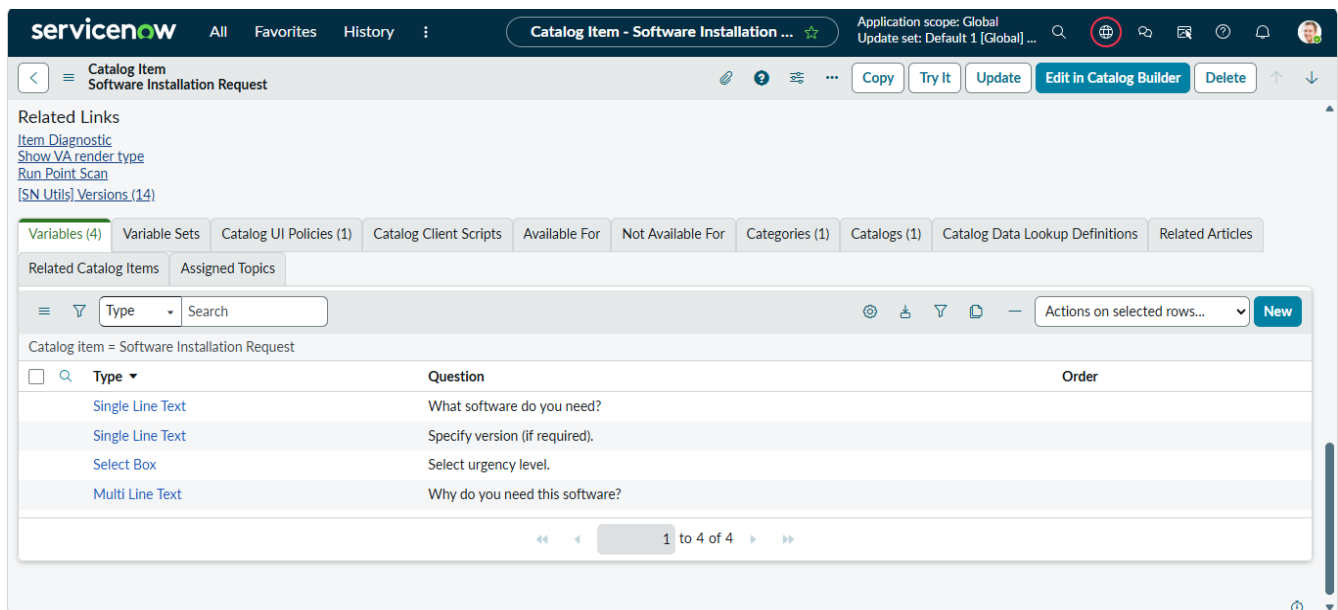
12.2 Dynamic Values

To reduce manual entry, values such as Requested For can be populated from the logged-in user. Department and Location can also be pulled dynamically from the user's profile when the corresponding configuration is implemented.

12.3 Tooltip and Help Text

Tooltip/help text can be added next to licensing-related information so that employees understand whether the requested software requires an available license or additional approval.

Catalog variables:



The screenshot displays the ServiceNow interface for a 'Catalog Item - Software Installation Request'. The 'Variables' tab is selected, showing a list of variables for the catalog item. The variables are organized into a table with columns for 'Type', 'Question', and 'Order'.

Type	Question	Order
Single Line Text	What software do you need?	
Single Line Text	Specify version (if required).	
Select Box	Select urgency level.	
Multi Line Text	Why do you need this software?	

The interface includes a search bar, a 'Type' dropdown, and a 'Search' button. The 'Catalog item = Software Installation Request' is noted at the top of the table. The bottom of the table shows a pagination indicator '1 to 4 of 4'.

13. Phase 3 – Navigation Flow

The navigation flow provides the employee with a simple path from the Service Portal to the Software Installation Request.

13.1 Procedure

1. Log in to the ServiceNow PDI.
2. Open the Service Portal.
3. Search for the Software Installation Request catalog item.
4. Open the item.
5. Enter the required software details.
6. Review the mandatory information.
7. Click Order Now.
8. Record the generated request number.
9. Track the request through the available request view and notifications.

13.2 User Flow

Service Portal → Software Installation Request → Enter Variables → Order Now → REQ/RITM Generated → Approval → Software Support Task → Fulfilment → Completion Notification

Service Portal Software Request:

The screenshot displays the ServiceNow user interface for a 'Software Installation Request'. The top navigation bar includes links for Knowledge, Catalog, Requests, System Status, Cart, Tours, and a user profile for 'System Administrator'. The breadcrumb trail shows the path: Home > All Catalogs > Service Catalog > Software > Software Installation Request. A search bar is located on the right. The main form area is titled 'Software Installation Request' with a subtitle 'Request to install company-approved software.' and a description 'Allows employees to request software installations. Fulfills via Software Support Team.' The form contains several input fields: 'What software do you need?' (labeled 'Software_Name') with the value 'Microsoft Office 365', 'Specify version (if required)' (labeled 'Version_required') with the value '2021', 'Why do you need this software?' (labeled 'License_Justification') with the value 'Required for daily work', and 'Select urgency level' (labeled 'Urgency') with a dropdown menu set to 'High | H'. On the right side of the form, there is a 'Quantity' field set to '1', a 'Delivery Time' of '1 Day', and three buttons: 'Add to Cart', 'Save as Draft', and 'Order Now'. At the bottom right of the form, there is an 'Add attachments' link. The bottom of the page features a chat icon on the left and a help icon on the right.

14. Phase 3 – Usability

Usability is addressed by presenting the required information in a single structured request form. Mandatory-field indicators, clear labels, help text and a simple Order Now action reduce confusion during submission.

14.1 Usability Features

- Single catalog item for software installation requests.
- Clear software name and version fields.
- Mandatory business justification.
- Automatic Requested For information where configured.
- Profile-based Department and Location values where configured.
- Tooltip/help text for licensing-related information.

14.2 Usability Validation

Test	Expected Result
Open catalog item	Software Installation Request is visible to authorized users.
Leave mandatory field blank	System prevents incomplete submission.
Enter valid data	Form accepts the request.
Submit request	REQ/RITM number is generated.
Open request	Submitted values are visible and traceable.
Check notification	Configured lifecycle notification is generated.

Usability / Order Number Output:

The screenshot displays the ServiceNow user interface. At the top, the navigation bar includes links for Knowledge, Catalog, Requests, System Status, Cart, Tours, and a user profile for System Administrator. The main header shows 'Home > Request Summary - REQ0010005' and a search bar labeled 'Search Catalog'. The request details are shown in a box: Submitted: 2026-08-28 09:13:41, Request Number: REQ0010005, and Estimated Delivery: 2026-08-30. Below this is a table with columns: Item, Delivery Date, Stage, Price (each), Quantity, and Total. The table contains one row for 'Software Installation Request' with a delivery date of 2026-08-30, a stage of 'Assess or Scope Task', and a quantity of 1. The total price is \$0.00. At the bottom left, there are icons for a pie chart and a globe. At the bottom right, there is a chat bubble icon.

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Software Installation Request	2026-08-30	Assess or Scope Task	---	1	---

Total: \$0.00

15. Phase 3 – UI Policies and Client-Side Logic

Catalog UI Policies and Catalog Client Scripts can control the behaviour of the Software Installation Request form. They help make the form responsive to user selections and reduce incorrect or unnecessary data entry.

15.1 UI Policy

A UI Policy can be configured to show, hide, make mandatory or make fields read-only based on conditions. For example, additional installation details can be displayed when a particular device or software option requires them.

15.2 Catalog Client Script

A Catalog Client Script can perform client-side actions when the catalog form loads or when a user changes a field. It can support controlled population or validation of values where appropriate.

15.3 Catalog Item Configuration Summary

Configuration	Purpose
Catalog Variables	Collect software request information
Variable Set	Group related variables
Catalog UI Policy	Control field behaviour dynamically
Catalog Client Script	Perform client-side actions
Catalog Availability	Control where and for whom the item is available

Catalog Item UI Policy / Client Script Configuration:

The screenshot shows the ServiceNow interface for configuring a Catalog UI Policy. The page title is 'Catalog UI Policy - Make Software V...'. The application scope is 'Global' and the update set is 'Default 1 [Global] ...'. The policy is named 'Catalog UI Policy - Make Software Version mandatory' and is currently 'Active'. The 'Applies to' field is set to 'A Catalog Item'. The 'Catalog item' field is set to 'Software Installation Request'. The 'Short description' is 'Make Software Version mandatory'. The 'When to Apply' tab is selected, showing a list of conditions: '1. The catalog UI policy is Active', '2. The items in the Conditions field evaluate to true', and '3. The field specified in the catalog UI policy is present on the specified catalog item'. There are buttons for 'Add Filter Condition' and 'Add OR Clause'. The 'Applies on a Catalog Item view' checkbox is checked, and the 'Applies on Catalog Tasks' checkbox is unchecked. A blue box at the bottom right states: 'Apply the catalog UI policy actions when the form is loaded or when the user changes values on the form'.

16. Phase 4 – Data Migration, Testing & Security

Phase 4 validates the automated workflow, protects the software request data through access controls, performs quality testing and confirms data integrity.

16.1 Automation Overview

The software request process is automated using ServiceNow Flow Designer. The flow uses the submitted catalog request, retrieves the relevant variables, initiates approval, handles approval outcomes, creates the fulfilment task and updates the request status.

16.2 Automation Sequence

1. Trigger when the Software Installation Request is submitted.
2. Read the Requested Item and catalog variables.
3. Route the request to the configured approval step.
4. If rejected, update the request according to the configured rejection path and notify the requester.
5. If approved, create a Catalog Task for Software Support.
6. Update task and request states as fulfilment progresses.
7. Generate completion notification after successful fulfilment.

Software Installation Request Flow Designer :

The screenshot displays the ServiceNow interface for configuring a 'Catalog Item - Software Installation Request'. The left sidebar shows the 'Service Catalog' and 'Catalog Definitions' sections. The main content area is titled 'Catalog Item - Software Installation Request' and includes a search bar, a 'Service Catalog' link, and a 'Fulfillment automation level' dropdown set to 'Unspecified'. The 'Process Engine' tab is selected, showing a configuration table with columns for 'Flow', 'Workflow', and 'Execution Plan'. The 'Flow' field is empty, 'Workflow' is set to 'Software Request Workflow', and 'Execution Plan' is empty. A message states: 'Select the appropriate process engine for the catalog Item. Only one engine can be selected.' Below the table are buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. The bottom section shows 'Related Links' for 'Item Diagnostic' and 'Run Point Scan'.

Flow	Workflow	Execution Plan
	Software Request Workflow.	

17. Phase 4 – Automation Logic in Flow Designer

17.1 Flow Designer Activities

Activity	Purpose	Output
Trigger	Starts the automation when the catalog request is submitted	Flow execution begins
Get / Read Request Data	Obtains RITM and variable information	Request data available
Approval	Routes request to approver	Approved / Rejected outcome
Decision / Branch	Handles approval outcomes	Correct lifecycle branch
Create Catalog Task	Creates fulfilment work	SCTASK assigned to Software Support
Update Record	Updates task/request status	Current lifecycle state
Notification	Communicates progress or completion	Email/message output

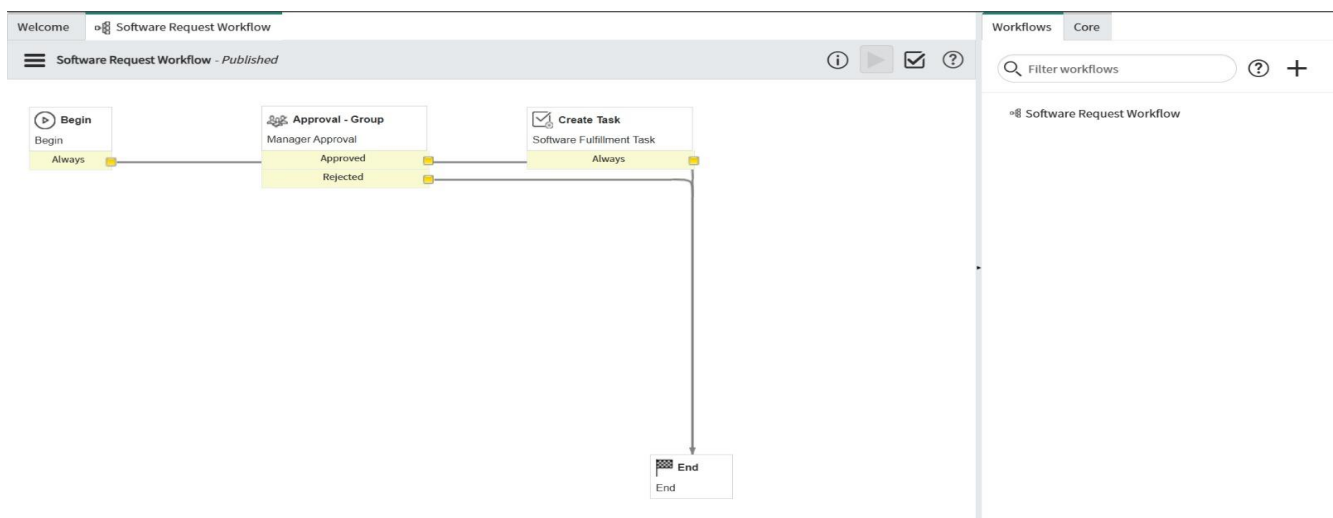
17.2 Approval Branch

The approval branch is a key control point. The request should not proceed to software installation until the configured approval condition is satisfied.

17.3 Rejection / Exception Branch

Rejected requests can be stopped, marked appropriately, and communicated to the requester. A license-shortage condition can also place the request on hold and trigger the configured exception handling.

Approval and update branches:



18. Phase 4 – Access Control and Security

Access control ensures that software request information is available only to authorized users and groups. ACL configurations can control read, create, write and delete operations for relevant records.

18.1 Access Control Procedure

1. Navigate to All → System Security → Access Control (ACL).
2. Identify ACLs relevant to the software request records.
3. Verify create, read and write permissions.
4. Ensure employees can create and view their permitted requests.
5. Ensure approvers have the required approval access.
6. Ensure Software Support users can work on fulfilment tasks.
7. Restrict unauthorized modifications to request data.

18.2 Security Principles

- Least-privilege access.
- Role-based record access.
- Controlled approval actions.
- Protection of request information.
- Traceable changes through ServiceNow records and workflow history.

ACL configuration:

The screenshot shows the ServiceNow interface for configuring an Access Control Rule (ACL) for the 'sc_req_item.cat_item' table. The page is titled 'Access Control - sc_req_item.cat_item' and includes a breadcrumb trail: 'All > Favorites > History > Access Control - sc_req_item.cat_item'. The configuration fields are as follows:

- Type:** record
- Operation:** read
- Decision Type:** Allow If
- Application:** Global
- Active:** ☒
- Advanced:** ☐
- Admin overrides:** ☒
- Protection policy:** -- None --
- Name:** sc_req_item.cat_item
- Description:** (empty text area)
- Applies To:** No. of records matching the condition: 17 (empty)

Below the configuration fields, there is a 'Conditions' section with a blue informational box stating:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

A 'More Info' link is provided at the bottom of the informational box.

19. Phase 4 – QA Testing

End-to-end QA testing validates the complete lifecycle from request submission through approval, task creation, fulfilment, status update and notification.

Test Case	Action	Expected Result
TC01	Open Service Portal and search Software Installation Request	Catalog item is displayed
TC02	Submit without required software name	Validation prevents submission
TC03	Submit valid software request	REQ/RITM is generated
TC04	Check Requested For	Correct requester is populated where configured
TC05	Verify approval	Approval record is generated
TC06	Approve request	Flow proceeds to fulfilment
TC07	Verify task	Catalog Task is created for Software Support
TC08	Complete fulfilment task	Task and request states update
TC09	Verify notification	Configured notification is generated
TC10	Check request history	Lifecycle is traceable

Successful request submission:

servicenow

AllFavoritesHistoryAdmin

Order Status: REQ0010007

Application scope: Global
Update set: Default 1 [Global]

Search

Order Status

Back to Catalog

Continue Shopping

Home

Thank you, your request has been submitted

Order Placed: 2026-08-29 09:15:18

Request Number: [REQ0010007](#)

Estimated Delivery Date of Complete Order: 2026-08-31

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
Request to install company-approved software.	2026-08-31			1	
				Total	-

Back to Catalog

Continue Shopping

Home

20. Phase 4 – Notification Testing

Notifications keep the requester and relevant stakeholders informed about important lifecycle events. The exact notification conditions should match the configured flow and business rules.

20.1 Notification Events

- Request submission confirmation.
- Approval request to the configured approver.
- Approval or rejection outcome where configured.
- Fulfilment task or progress notification where configured.
- Completion notification to the requester.
- Exception notification when the request cannot be fulfilled due to a configured condition.

20.2 Notification Validation

1. Submit a valid software installation request.
2. Verify the request confirmation output.
3. Verify that the configured approver receives the approval notification.
4. Approve or reject the request.
5. Verify the corresponding lifecycle notification.
6. Complete the fulfilment task.
7. Verify the completion notification.

Email notification output:

The screenshot shows the ServiceNow interface for configuring an email notification. The page title is "Email - Software Installation Request Submitted". The configuration fields are as follows:

- Type:** send-ready
- Target:** Requested Item: RITM0010010
- User:** (empty field)
- Notification type:** SMTP
- UID:** N/A
- Created:** 2026-08-29 23:19:53
- Deleted:** ☐
- Weight:** 0
- Importance:** (empty field)

The "Originating Event and Notification" section shows:

Event	Notification
notification_engine.process	Software Installation Request Notificati

The email content fields are:

- Subject:** Software Installation Request Submitted
- Recipients:** jadmin@example.com
- Body:**

```
<html><head><style>
#snc_notification_preference {
font-size: 10px;
}
```

21. Phase 4 – Data Integrity

Data integrity is maintained by ensuring that required values are captured, automatically populated information is correct, approvals are respected, and relationships between REQ, RITM, approval and task records remain traceable.

21.1 Field Validation and Automation

- Software Name, Version and Business Justification are treated as mandatory information.
- Requested For can default to the logged-in user.
- Department and Location can be retrieved from the user profile where configured.
- Approval state is checked before fulfilment proceeds.
- Submitted catalog variables remain available on the related RITM.

21.2 Data Integrity Test Procedure

1. Open the generated Software Installation Request record.
2. Verify all submitted values are stored correctly.
3. Check the relationship between REQ and RITM.
4. Verify approval information is associated with the request.
5. Verify the Catalog Task references the requested item.
6. Confirm that no required information is missing.
7. Confirm that the lifecycle is recorded correctly.

22. Phase 4 – Data Integrity Matrix

Area	Actions Taken
Data Handling	Catalog variables are mapped to the request lifecycle and used by the automation.
Access Control	Role-based access and approval logic are applied through ServiceNow security configuration.
QA Testing	Full-cycle testing covers request, approval, task and notification outputs.
Data Integrity	Mandatory validation, auto-population and approval-state checks are applied.
Traceability	REQ, RITM, approval and SCTASK records provide lifecycle references.
Exception Handling	License shortage / On Hold handling can trigger the configured support process.

22.1 Expected Data Relationships

The intended relationship is: Catalog Item → REQ → RITM → Approval → SCTASK → Completion / Notification.

Data relationship evidence:

The screenshot displays the ServiceNow interface for a Requested Item (RITM0000001). The top navigation bar includes 'All', 'Favorites', 'History', and 'Admin'. The main header shows the record ID and a star icon. The 'Activities' section lists the user 'System Administrator' and the record's status: Impact 3 - Low, Opened by System Administrator, Priority 4 - Low, and State Open. Below this, the 'Catalog Tasks (1)' tab is active, showing a table with one task. The task is assigned to 'Procurement' and has a short description 'Order from vendor or move from in-stock inventory'. The table also includes columns for 'Actual start' and 'Actual end', both of which are empty. The bottom of the page shows a pagination bar indicating '1 to 1 of 1'.

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
TASK0000001	Procurement	(empty)	Order from vendor or move from in-stock inventory	(empty)	(empty)	

23. Phase 5 – Deployment, Documentation & Final Presentation

Phase 5 is the final implementation, documentation and demonstration stage. It consolidates the working configuration and prepares evidence for evaluation.

23.1 Troubleshooting

1. Review the complete Software Installation Request flow.
2. Check catalog item, variables and UI logic.
3. Check approval configuration and approver assignment.
4. Verify Catalog Task creation and assignment.
5. Check notification conditions and recipients.
6. Review Flow Designer execution details for errors.
7. Perform final end-to-end verification after corrections.

23.2 Adherence to Timelines

- Review the planned five project phases.
- Verify all configured components are completed.
- Complete testing and deployment within the planned schedule.
- Track pending issues and finish corrections before final demonstration.

23.3 Innovation

- Automation replaces manual email-based coordination.
- Approval routing provides structured authorization.
- Catalog Task creation reduces manual assignment.
- Request tracking improves transparency.
- License-shortage exception handling supports operational follow-up.

24. Phase 5 – Functional Documentation

24.1 Functional Overview

The system provides an employee-facing Service Catalog item for requesting licensed software. The employee enters software details and business justification, submits the request, and receives a request reference. ServiceNow then performs the configured approval and fulfilment lifecycle.

24.2 Technical Blueprint

Component	Configuration / Purpose
Service Portal	Employee-facing access point
Catalog Item	Software Installation Request
Catalog Variables	Capture software and requester information
Variable Set	Group related variables
REQ / RITM	Store request transaction
Approval	Authorize software installation
Flow Designer	Automate lifecycle
Catalog Task	Assign installation work
Business Rules	Automate backend conditions and exception logic
Notifications	Communicate lifecycle events
ACLs	Restrict data access
Update Set	Move configuration between instances

24.3 Setup Manual – High-Level

1. Create the catalog item.
2. Create and configure catalog variables.
3. Create or associate a variable set.
4. Configure UI policies/client scripts as required.
5. Configure approval logic.
6. Build the Flow Designer automation.
7. Configure fulfilment task assignment.
8. Configure notifications.
9. Configure relevant ACLs.
10. Test the complete lifecycle.

25. Phase 5 – Update Set and Deployment

Update Sets are used to move configured ServiceNow changes from one instance to another. The project configuration can be exported and retrieved in a test instance for validation.

25.1 Finalize the Update Set

1. Open the Update Set created for the project.
2. Review the captured configuration records.
3. Change the state from In Progress to Complete.
4. Click Update.
5. Use Export to XML to download the update set file.

25.2 Test the Update Set in Another Instance

1. Open the target ServiceNow instance.
2. Search for Retrieve Update Sets.
3. Choose the option to import an Update Set from XML.
4. Upload the exported file.
5. Open the imported update set.
6. Run Preview Update Set.
7. Review any preview errors or missing elements.
8. Commit the Update Set after successful validation.
9. Validate the transferred catalog, variables, flow and related configuration.

Update Set export / retrieval:

The screenshot displays the ServiceNow interface for an 'Update Set - Software Installation Request Setup'. The page includes fields for Name, State (set to 'Complete'), Parent, Release date, Install date (2026-08-20 03:11:32), and Installed from. Below these fields are buttons for 'Update', 'Back Out', and 'Delete'. A 'Related Links' section provides links for 'Export to XML', 'Merge With Another Update Set', 'Scan Update Set', and '[SN Utils] Versions (0)'. At the bottom, there are tabs for 'Customer Updates', 'Update Set Logs (3)', 'Child Update Sets', and 'Install History'.

Overlaid on the right side of the screenshot is a Windows 'Downloads' folder window. It contains a list of files, each with a green icon and a blue 'Open file' link below it. The files are:

- sys_remote_update_set_c81396ac838b4b503d92b5dfeead3b1.xml
- sys_remote_update_set_37f29e6c838b4b503d92b5dfeead3cd.xml
- sys_remote_update_set_ebd29e6c838b4b503d92b5dfeead3c7.xml
- sys_remote_update_set_57c25e6c838b4b503d92b5dfeead3c8.xml
- Software_Installation_Request_Automation_ServiceNow_Documentation.docx

A 'See more' link is visible at the bottom of the Downloads window.

26. End-to-End Functional Output

The completed solution follows a connected lifecycle from employee request to fulfilment and notification.

26.1 End-to-End Sequence

1. Employee opens the Software Installation Request through the Service Portal.
2. Requester enters software name, version, business justification and other required information.
3. The user selects Order Now.
4. ServiceNow generates a Request (REQ) and Requested Item (RITM).
5. The configured automation retrieves the catalog variables.
6. An approval request is generated for the configured approver.
7. The approver reviews and approves or rejects the request.
8. When approved, a Catalog Task is created for the Software Support Team.
9. The fulfilment team processes the task and updates its state.
10. The request status is updated as the task progresses.
11. After successful completion, a completion notification is generated.

26.2 Final Workflow

Service Portal → Software Installation Request → REQ/RITM → Approval → Software Support Catalog Task → Installation → Task Completion → Status Update → Email Notification

End-to-end request submission output:

The screenshot displays the ServiceNow interface for a Catalog Task. The top navigation bar includes the ServiceNow logo, tabs for All, Favorites, History, and Admin, and a search bar. The main header shows the task title 'Catalog Task - SCTASK0010010' and the application scope 'Global'. The form is divided into two columns. The left column contains fields for 'Number' (SCTASK0010010), 'Assigned to' (with a search icon), 'Assignment group' (Software, with a search icon and an information icon), 'Configuration item' (with a search icon), 'Active' (checkbox), 'Short description' (Provide requested service, with a search icon and an information icon), 'Description' (text area), and 'Work notes' (text area with a 'Post' button). The right column contains fields for 'Approval' (Not Yet Requested), 'Priority' (4 - Low), 'State' (Closed Complete), 'Request item' (RITM0010010, with a search icon and an information icon), and 'Requested for' (System Administrator, with a search icon and an information icon). At the bottom, the 'Activities' section shows 4 activities, with the first activity being 'System Administrator' and the state 'Closed Complete was Work in Progress'. A 'Field changes' section shows a change on 2024-08-29 23:28:00.

Field	Value
Number	SCTASK0010010
Assigned to	
Assignment group	Software
Configuration item	
Active	☐
Short description	Provide requested service
Description	
Work notes	Work notes
Approval	Not Yet Requested
Priority	4 - Low
State	Closed Complete
Request item	RITM0010010
Requested for	System Administrator

Activities: 4

System Administrator

State: Closed Complete was Work in Progress

Field changes: 2024-08-29 23:28:00

27. Project Demo Video Planning and Visual Demonstration

27.1 Recommended Demo Sequence

1. Introduce the business problem and project objective.
2. Open the ServiceNow Service Portal.
3. Search for the Software Installation Request.
4. Show the catalog item and required variables.
5. Enter sample software information and business justification.
6. Submit using Order Now.
7. Show the generated REQ and RITM.
8. Show the approval record and approval decision.
9. Show the automatically created Software Support Catalog Task.
10. Update the task to demonstrate fulfilment.
11. Show the final request status.
12. Show the completion notification.

27.2 Visual Evidence Checklist

- Catalog item configuration.
- Catalog variables.
- Service Portal request form.
- Generated REQ/RITM.
- Approval record.
- Flow Designer automation.
- Catalog Task assigned to Software Support.
- Notification/email output.
- Update Set deployment evidence.

27.3 Explanation Clarity

During the presentation, explain the purpose of each major ServiceNow component and show how the records move from one stage to the next. Keep the demonstration aligned with the actual configuration and screenshots.

28. Realism, Quality, Scalability and Future Plan

28.1 Realism & Quality

- Demonstrate the actual configured ServiceNow system.
- Use screenshots and records that represent the implemented configuration.
- Verify that the workflow produces the expected request, approval and task outputs.
- Use consistent terminology across the documentation and demonstration.
- Validate notifications and lifecycle status before final presentation.

28.2 Scalability

The software installation solution can be extended to additional software categories and approval policies without changing the overall lifecycle model.

- Add more licensed software catalog items or software categories.
- Introduce multiple approval levels for higher-risk applications.
- Add security approval for restricted applications.
- Integrate license inventory or software asset information in future versions.
- Add automated installation integrations where supported.
- Expand reporting and monitoring dashboards.

28.3 Future Enhancements

- Automated license availability checks before approval.
- Integration with endpoint management tools for installation.
- SLA tracking and escalation.
- Automated reminders for pending approvals.
- Analytics for request volume, turnaround time and rejection reasons.
- Knowledge articles and self-service recommendations.

29. Project Conclusion

The Software Installation Request Automation in ServiceNow project implements a structured and automated process for employees requesting licensed software. The solution connects Service Catalog, catalog variables, Request and Requested Item records, approvals, Flow Designer, Catalog Tasks, Business Rules, UI logic, notifications, access controls and Update Sets into a single lifecycle.

The system reduces manual intervention by standardizing request intake and automatically moving approved requests toward fulfilment. It improves visibility by providing request numbers, related records and lifecycle status. Mandatory validation and controlled approvals help maintain data quality and support licensing, governance and security requirements.

The final workflow is: Request Submission → Request Creation → Approval → Software Support Task → Fulfilment → Status Update → Notification.

29.1 Key Outcomes

- Centralized licensed-software request submission.
- Structured software and business-justification information.
- Automated approval process.
- Automated Software Support task creation.
- Improved request tracking and visibility.
- Reduced manual coordination.
- Status updates throughout the request lifecycle.
- Completion notification to the requester.
- Better support for licensing and governance controls.

29.2 Final Statement

The project demonstrates how ServiceNow can be used to convert a manual software installation request process into a controlled, transparent and scalable service workflow. The documented configuration, testing and deployment approach provides a foundation for future expansion into broader software asset and IT service management processes.